IGBA'S "Shooting for the Stars" Challenge

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Story

The Intergalactic Basketball Association (IGBA) is the official basketball league of the solar system. Each planet within our solar system, with the exception of Earth, is represented by the best representatives that each planet has to offer. For years, Earth has passed on inclusion in the IGBA due to technological constraints that prevent humans from meeting the travel, health and safety demands required to participate in the IGBA.

The current landscape of the IGBA consists of 9 teams across 2 conferences. Each team and the conference they belong to is shown below.

Terrestrial Conference	Jovian Conference
Sun Flares	Jupiter Giants
Mercury Messengers	Saturn Golden Rings
Venus Magmas	Uranus Icicles
Mars' Reds	Neptune Retros
	Pluto Dwarfs

What makes this league unique is the differences amongst the environmental factors of each planet. All planets vary on factors such as temperature, gravity, weather conditions, and overall livability. These factors give all new meaning to "home field advantage". Teams established within the IGBA must be capable of interplanetary travel and adaptation to the harsh environments of their opponents, all within the time constraints of the basketball season.

In recent years, the IGBA has been looking for a solution to problems with the current situation in the league. Having an uneven number of teams creates inefficiencies with scheduling, which has resulted in a decrease in revenue. At this point in time, they have developed two options…

1.) ***Remove the Pluto Dwarfs from the IGBA***. This creates an even number of teams in the league and maximizes efficiency by removing the smallest market and furthest planet.
2.) ***Recruit Earth to join the IGBA***. Earth's space technology has shown potential in the last decade. With some assistance, Earth could be capable of participating in the league sooner than was expected.

All involved parties of the IGBA would prefer option 2, but it is clearly the more difficult option. If it were to work according to plan, the IGBA would be able to tap into the immense popularity of basketball on Earth and increase their market share and revenue. The question now is how will this be done?

The Pluto Dwarfs have taken a significant interest in bringing Earth onboard to the IGBA, being that they will be disbanded if not. Being that Pluto is the smallest planet, they have the smallest market share and to remove that team from the IGBA would be of the least impact to the league overall. But Pluto can't do this by themselves. Earth's team is in need of a financial backer, someone who is very knowledgeable in all matters regarding our solar system, and someone who knows the game of basketball. For this role, the Pluto Dwarfs has partnered with Atom Silver, world renowned astrophysicist, multi-billionaire businessman, and the greatest NBA player to ever live. Silver's first love was the game of basketball, but he always took an interest in all things outer space-related. When he was young, Silver used to shoot hoops in his backyard for hours at a time. He would stay up until very late at night just to "make one more basket" and stare up at the stars in between shots. When he grew up, Silver went on to play professionally and had a heavily decorated career. Because of his passion for outer space, Silver received a doctorate in astrophysics and has been developing technology with his own company to search the solar system for sustainable habitats. He is the perfect person to back such a venture.

Working in association with the teams of the IGBA, the Pluto Dwarfs and Atom Silver have developed an intergalactic tour to recruit human talent to play for Earth's new team, the Earth Citizens. You and your party have been invited to try out for this team.

This tour starts where it all began for Atom Silver. Recruits are welcomed into Silver's childhood house where all information is recorded before beginning the outer space journey. Guests then go to the backyard; this area is an asphalt driveway with a basketball hoop. This is where Silver used to spend this childhood days shooting hoops. Guests can warm up here while gazing at the planetarium-like night sky. After a short while, your ship arrives. From there, it is off to the IGBA Space Hub, where you will be able to transport to other planets and take part in this galactic mission!

Game Rules

Game: *IGBA's "Shooting for the Stars" Challenge*

Overall Rules:

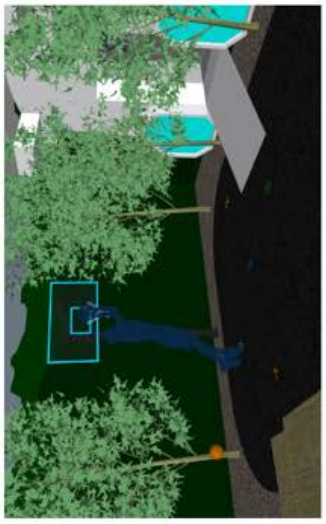
- A guest can choose which numbered spot they wish to attempt a shot from. Spots to shoot from range from #1 to #5. The difficulty level increases as the numbers go up, but with that so does the point value.
- The guest has 3 attempts to shoot and sink the basket. For each made basket, the guest will receive the point value associated with the designated spot they shot from. If the shot is not made, no points will be awarded for that attempt. Guests can adjust shooting spots for each attempt.
- Whoever has the highest score by the end of the game wins the challenge and will be considered to play for the Earth Citizens.
- For junior candidates, the hoop can be lowered and point values adjusted to make the competition fair for participants of all ages.
- Using RFID technology, sensors in their sweatband gear handed out at check in will register their location and calculate their total points.

Game Concept: The IGBA is considering Pluto's request of expanding the league to Earth. To do so, talent from Earth needs to be recruited and assessed to form a professional team. With representatives from the Pluto Dwarfs as your guide, your party will travel to each planet and compete against one another to put up the highest score. Whoever has the highest score by the end of the game wins and will be considered to play for the Earth Citizens.

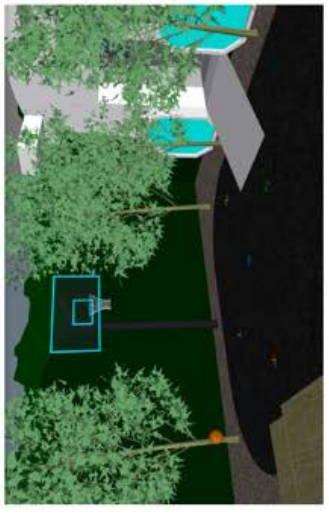
Storyboard Displaying Game Play



1. There are five spots from which a player can shoot a basketball. They are labeled from 1, which is the easiest spot, to 5, which is the hardest spot. You get three attempts to shoot per room. Each time you score, you will get the same number of points as the spot you are standing on. These are the spots in Atom Silver's backyard.

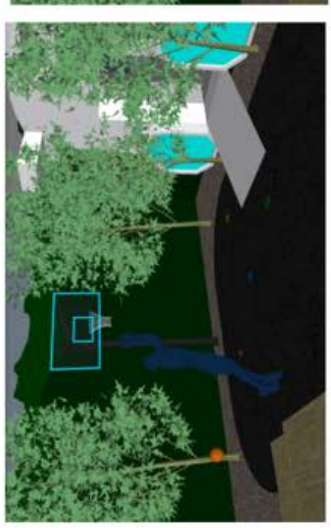


2. In this image, the guest is shooting from the number 1 spot, the easiest position. If he sinks the basket, then he will be awarded one point.

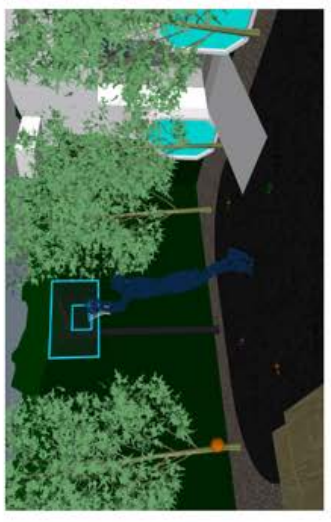


4. In this image, the guest is shooting from the number 3 spot, the third most difficult position. If he sinks the basket, then he will be awarded three points.

3. In this image, the guest is shooting from the number 2 spot, the second easiest spot. If he sinks the basket, then he will be awarded two points.



5. In this image, the guest is shooting from the number 4 spot, the second most difficult spot. If he sinks the basket, then he will be awarded four points.



6. In this image, the guest is shooting from the number 5 spot, the most difficult position. If he sinks the basket, then he will be awarded five points.

Note: The silhouette basketball player was created by vexels.com (https://www.vexels.com/png-svg/preview/130628/glossy-basketball)

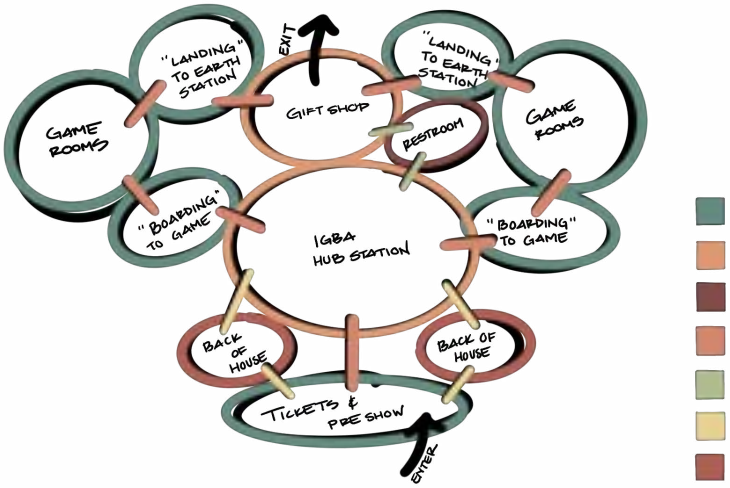
User Journey

Scene	Name
1	Atom Silver's House
2	Silver's Backyard
3	Space Shuttle
4	IGBA Hub
5	Space Transport
6	Pluto Dwarf's Stadium
7	Space Transport
8	Mars Red's Stadium
9	Space Transport
10	Jupiter Giants Stadium
11	Space Transport
12	Saturn Golden Ring's Stadium
13	Space Transport
14	Uranus Icicle's Stadium
15	Space Transport
16	Neptune Retro's Stadium
17	Space Transport
18	Venus Magma's Stadium
19	Space Transport
20	Mercury Messenger's Stadium
21	Space Transport
22	Sun Flare's Stadium
23	Space Transport
24	Earth Citizen's New Stadium
25	Space Transport
26	Earth's IGBA Station
27	Gift Shop

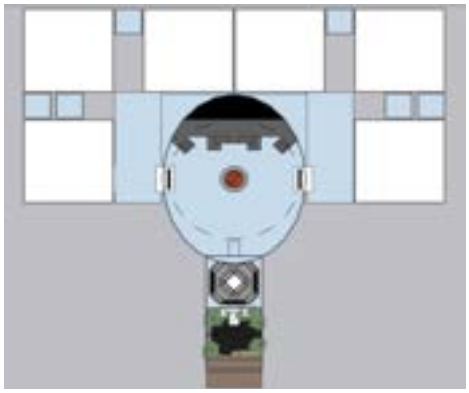
The Overall Space

The overall design of this game can be seen in Figure (OS.2), where you will see Atom Silver's house leading into his backyard. After you make your way through his backyard, you will enter a Spaceship to the Intergalactic Basketball Association (IGBA) Hub! To begin the game, you will take an elevator to one of the game rooms that are displayed on the left and the right of the circular structure in Figure (OS.2). A series of elevator rides will take you to and from different game rooms until finally, you have finished the game and are dumped into a recreation of Atom Silver's backyard at the back of the game building. There will be an area for guests to shoot hoops in this room and exit through the gift shop, which is a recreation of Atom Silver's house. Both of these recreations are different from the original backyard and house that are included at the beginning of the whole experience. This overall user journey can also be seen in Figure (OS.1)

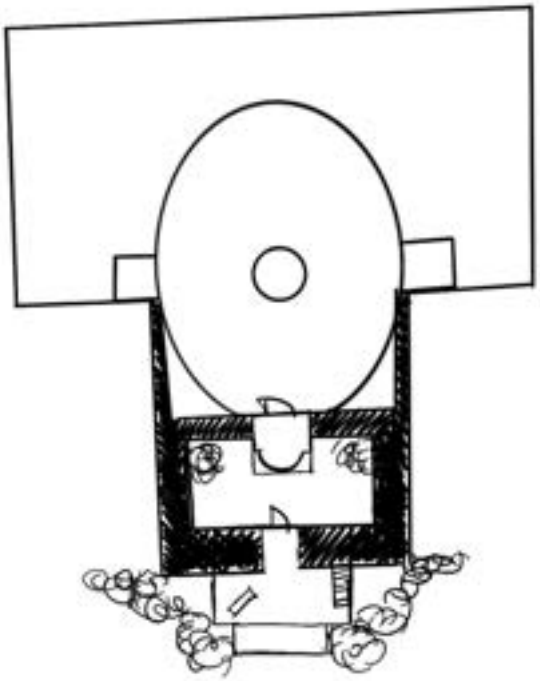
(OS.1)

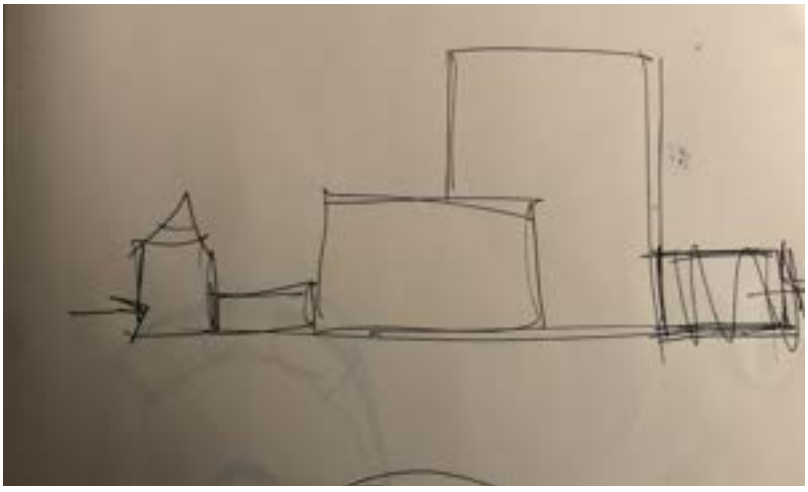


(OS.2)



Rough Sketches of the Overall Place





Earth: Inside Atom Silver's House

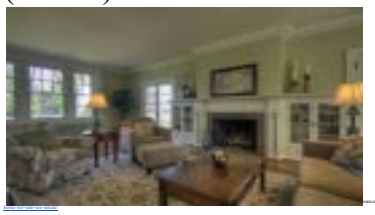
When you enter Atom Silver's house, you walk straight into his living room, which has heavy inspiration drawn from Figures (ASH.2 and ASH.3). There is a couch on the right, and two seats on the left sitting on a rug. They are all slanted, facing the center of the room where a fireplace is located. Next to the fireplace on the left is where a door to the backyard can be found. In the center of the room, in between the couches and chairs, there is a big brown table where the employees of the game will be distributing tickets and the sweatbands to keep track of points. All around the house are pictures of famous basketball players, such as Lebron James, Michael Jordan, Magic Johnson, Shaq, and more! Pictures of Atom Silver and his family are put up around the walls and standing on tables, and many of his basketball trophies are showcased in a trophy case on the right of the fireplace. Above the fireplace, you can see a television that is playing all of Atom Silver's most famous plays! The walls of the room are painted a beige-green, and sitting around the room are some of Atom Silver's astrophysicist inventions, including the first prototype models for the transportation device that he has created to bring guests to space. Similarly, newspapers and articles about the IGBA are just scattered throughout the house!

(ASH.1)



Source:https://www.metrotimes.com/city-slang/archives/2017/06/27/you-can-purchase-madonnas-rochester-hills-childhood-home

(ASH.2)



(ASH.3)



Earth: Atom Silver's Backyard

Walking out of Atom Silver's house, you see a big basketball court in the backyard with a blacktop similar to the one seen in Figure (AB.1). On the left of the fenced in backyard, you see the basketball hoop standing tall. On the blacktop of the court, you see five different spots marked out in chalk to show where you can practice shooting your basketball. Surrounding the blacktop basketball court is a whole bunch of grass and some dried up leaves. When you look to the right, you see a projector screen and some lawn chairs like what is shown in Figure (AB.4). This is where you will see a preshow for the whole experience! On the far back side of the room, you see a big white structure that appears like a teleportation device. This is the device that Atom Silver built to bring you to space. Surrounding the entire room are many different trees reaching up to the light blue sky. The temperature in this room is 70 degrees Fahrenheit, and the weight of a basketball on this planet is the exact average weight of a ball on a planet with a gravity of one, which is 1.5 pounds! In the sky, you can seem some clouds and the Sun far away blaring down on you.

(AB.1)



Source:https://www.indiegogo.com/projects/home-court-advantage-diy-basketball-stencil

(AB.3)



Source:backyard basketball court

(AB.2)



Source: https://www.residencestyle.com/5-reasons-to-add-a-basketball-court-to-your-backyard/

(AB.4)



Source:https://www.hgtv.com/design/make-and-celebrate/handmade/outdoor-movie-screen

The Spaceship to the IGBA Hub

Inside the Spaceship to the IGBA Hub, when you walk in, you will see a big structure in the middle of the room that is square and transparent, allowing you to look down into the space that you are traveling over. However, once the ride begins, a holographic projection begins to show in between the white panels of this structure shown in Figure (SS.1). This projection is a video preparing you for the journey you are about to embark on! Surrounding the room you notice many control panels all around that are responsible for helping your journey up into space. There are also windows surrounding the elevator so that you may look outside as you fly into the sky. The room itself is relatively grey, with hints of yellow, orange, blue, and black. The Figure below shows exactly what this room should look like!

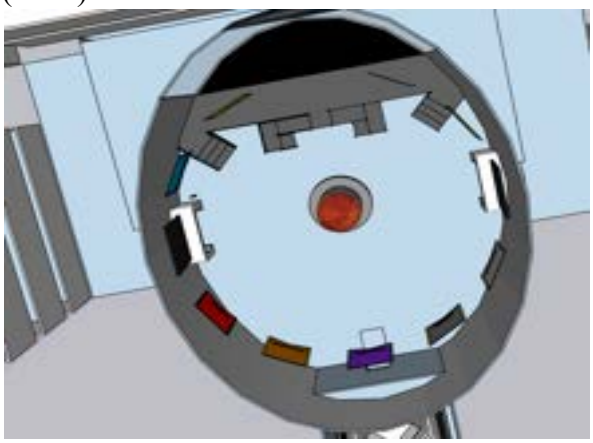
(SS.1)



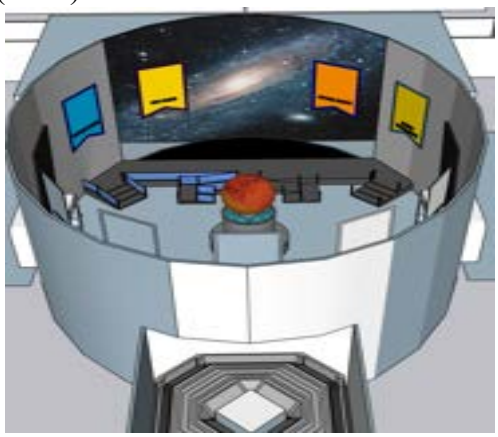
The IGBA Space Hub

As you step off of the Spaceship to the IGBA Space Hub and enter the Hub itself, you enter a huge building with silver/grey walls surrounding a metallic blue floor seen in the figures below. Around these walls are pennants of all of the different teams currently in the IGBA, including the Mercury Messengers, Saturn Golden Rings, Uranus Icicles, Jupiter Giants, Sun Flares, Neptune Retros, Mars Reds, Venus Magmas, and Pluto Dwarfs. At the back of the room there is a stage where performances or award ceremonies will take place for the different tournaments, but above the space is an empty pennant that will eventually have Earth's team written on it once you defeat the "Shooting for the Stars" Challenge! Behind the stage, there is a huge glass window that shows the abyss of space! In the middle of the room is a fountain like the one seen in Figure (SH.4) that has a spinning planet on it which projects images of different planets onto each of its hemispheres! To the right and the left of the room are the entrances through which you may start your game! Both sides have identical structures. In order to start the game, though, you must choose one side and go through the elevator transportation advice to get to the next planet!

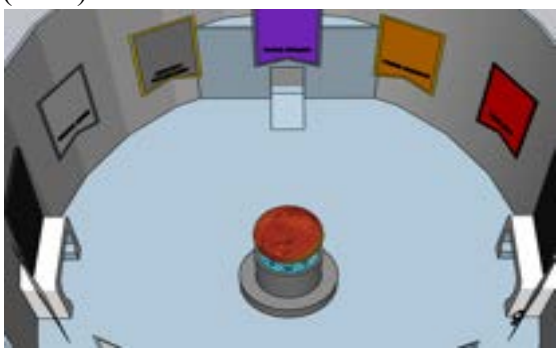
(SH.1)



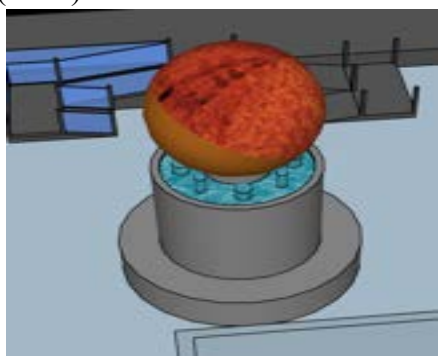
(SH.3)



(SH.2)



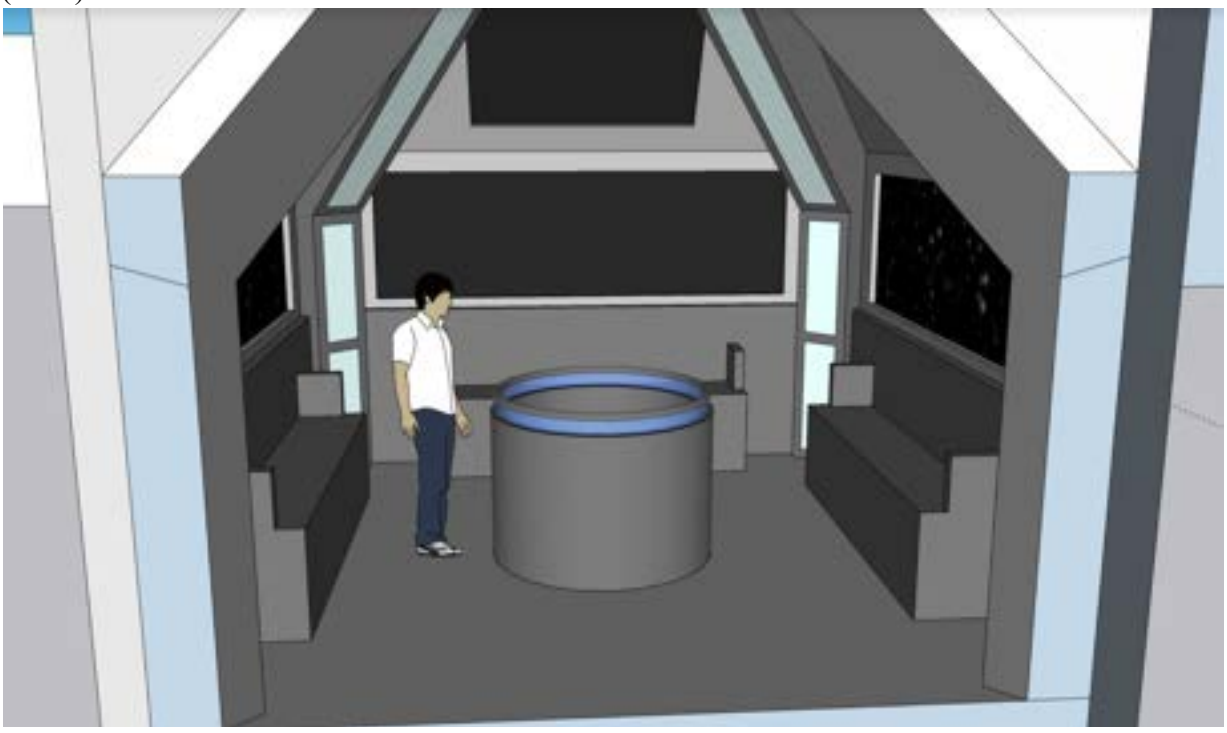
(SH.4)



Elevator Transports

Entering into the Elevator Transport, which will look the same for pretty much any planetary transportation in the game, you will see a square room that has a triangular structure on the top like the one seen in Figure (ET.1). The elevator is grey pretty much all around, and there are four benches in it, one on each side of the square. The middle of the room has a cylindrical structure that allows guests to look down and see the space they are flying over. Similarly, windows around the cabin allow for vision of space as well.

(ET.1)



Pluto

As you walk onto the planet Pluto, you are greeted by a very cold temperature of 58 degrees Fahrenheit and a moist atmosphere. The ground is composed of a dark brown and reddish rocky terrain with what appears to be ice projected on top of the entire floor. Figure (P.4) below will serve as a good reference image for this. In the back of the room, you will see a large rock tower similar to the one on the right in Figure (P.1). In the middle of the floor there are five spots of differing difficulty levels where game players stand to shoot their basket. Once the first basket of the game is shot, the ice projected onto the terrain immediately disappears and becomes fog, representing the tendency of ice to immediately turn into gas without first turning into a liquid on Pluto. If you look at the left of the room, you will see a big projection of Pluto's moon Charon, which is always visible on Pluto since it is stuck in the same orbit around the sun, making it tidally locked. The image of Charon is a big silver, grey moon with a large red spot facing the planet's surface. The surrounding sky is black with lots of white stars and a very dim sun covered by fog at the right side of the building. The surrounding sides of the room are also lined with brown, rocky mountains similar to the ones contained in the figures below. At the very back of the room, in front of the big rock structure that is similar to the one on the right of Figure (P.1), an adjustable basketball hoop will be located. The weight of basketballs on Pluto will be about .9 pounds to allow guests to feel the difference in gravity as they enter the new planet.

(P.1) 
Source:https://lightsinthedark.com/2015/06/05/how-bright-is-daytime-on-pluto/

(P.3) 
Source:https://iaaa.org/a-landscape-on-pluto/

(P.2) 
Source:https://griffithobservatory.org/exhibits/gunther-depths-of-space/the-planets/pluto-and-beyond/

(P.4) 
Source:https://www.space.com/29885-pluto-atmosphere-to-be-revealed-by-nasa-new-horizons-spacecraft.html

Mars

Entering onto the planet Mars, you feel a temperature of 68 degrees Fahrenheit. As soon as you step onto the planet, winds of red dust fill the air. The floor is reddish, orange rock full of small craters like the ones seen in Figure (Ma.4). To the left of the room, you will see half of the lake that used to exist in the Jezero Crater seen in Figure (Ma.1). It is filled with water to represent that there once was this beautiful lake on the surface of Mars. On the right side of the building is an exact replica of the Mars Rover. The Mars Rover is an animatronic that moves around in a circle and transmits messages from Earth alerting the team of how they are doing in the current competition of the IGBA. In the far back of the room, a big dusty tornado can be seen, as tornadoes are quite commonly found on Mars due to the dusty winds that inhabit the planet. Per game regulations, five spots will be located on the floor of the room to shoot the basketball from. You will be shooting into a hoop that is located right in the center of the Jezero Crater's lake! All around the room's floors, craters, remnants of tiny canyons, hills, and volcanic lava scatter the floor. The sky is a tannish brown with a few clouds similar to the sky photographed in Figure (Ma.3). Two moons can be seen in the sky, and they are Mars's moons of Pobos and Deimos. Since a normal basketball on Earth weighs about 1.5 pounds, taking into account the gravity of Mars, the basketballs provided on this planet will weigh about 1 pound.

(Ma.1)



Source:https://www.nasa.gov/image-feature/jezero-crater-was-a-lake-in-mars-ancient-past/

(Ma.3)



(Ma.2)



(Ma.4)



Source:https://nouse.co.uk/2019/10/29/human-mars-landing-so-much-more-than-a-small-step

Jupiter

In this game, the planet of Jupiter will feel like it is 66 degrees Fahrenheit. Jupiter is technically a gaseous planet, so it would not make sense that a person would be able to stand on it. However, for this game, the entire room is going to be created out of projected screens that project gaseous designs of red, orange, yellow, light blue, white, and tan colors. The projected gases are supposed to look similar to ideas conveyed in Figures (J.2) and (J.4). Since the part of the Jupiter this game is supposed to take place on is the Great Red Spot, the big red spot seen in Figure (J.1) will be at the center of the floor. Seventy nine moons will be swirling around the top of the ceiling to represent the seventy nine moons surrounding Jupiter. The entire room will be filled with foggy mist and the colors mentioned above. You are supposed to feel like you are walking in the clouds of Jupiter. The weight of the basketball will be about 1.7 pounds, and the basketball hoop will be in the far back right corner of the room.

(J.1)



Source:https://www.forbes.com/sites/jamiecartereurope/2021/08/17/whats-that-bright-star-in-the-east-why-the-king-of-the-planets-will-look-its-brightest-biggest-and-best-this-week/?sh=7146615e2a01

(J.3)



Source: https://nautil.us/blog/give-thanks-to-jupiter-our-little-planets-big-protector

(J.2)



Source:https://deepstash.com/idea/39057/pioneer

(J.4)

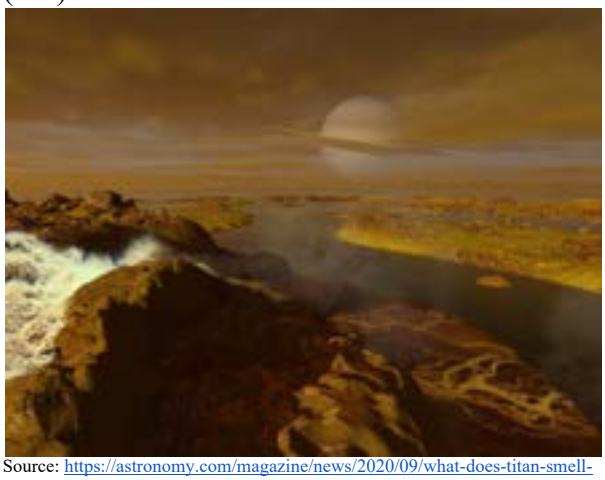


Source:https://www.pinterest.fr/pin/251849804135943360/

Saturn (Titan)

Saturn's moon Titan is a part of the galaxy that could potentially support life! For this reason, when you walk in, you will be greeted by a luscious field of grass surrounded by mountainous rocks, similar to Figure (T.1). The air has a slight fog to it, and the surrounding sky is a mixture of browns, blues, and oranges. To the right of the room, you will see the rings of Saturn jutting into the room itself, followed by the body of Saturn seen in a far away distance. Five spots on the floor represent the different areas where you can shoot your basketball. Caddy-cornered in the part of the room right in front of you as you enter is a pool of water hidden under a cave of rock. The temperature of this room is about 64 degrees Fahrenheit, and when accounting for gravity, the weight of a basketball on this planet is about 1.4 pounds.

(T.1)



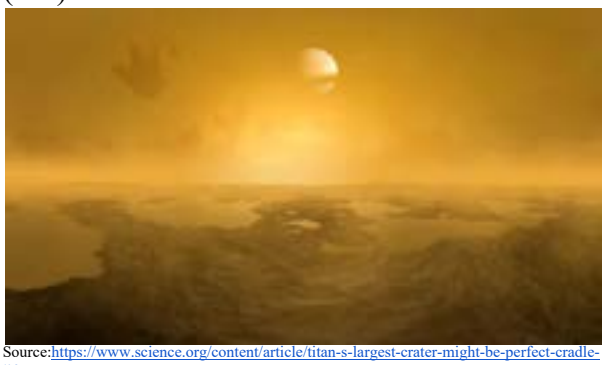
Source: https://astronomy.com/magazine/news/2020/09/what-does-titan-smell-like

(T.3)



Source: https://www.airspacemag.com/daily-planet/saturns-moon-titan-living-cells-may-be-very-different-ours-180974104/

(T.2)



Source:https://www.science.org/content/article/titan-s-largest-crater-might-be-perfect-cradle-life

(T.4)



Source:https://todayheadline.co/moon-in-a-jar-recreates-the-hazy-atmosphere-of-titan-saturns-largest-moon-livescience-com/

Rough Sketches of Titan



3D Model of Titan



Uranus

Similar to Jupiter, Uranus is completely gaseous and could technically not be inhabited nor stepped on by humans. However, for this game, you will walk in a room made out of projections of blue, green, and white gas. The whole room will be full of blue, green, and white clouds that make it a little difficult for you to see, similar to the gases and atmosphere seen in Figures (U.1) and (U.3). At the top of the room, twenty seven moons will orbit around. Surrounding the room, you will see rings of asteroids and clumps of dust orbiting the planet, representing the thirteen rings of Uranus seen in Figure (U.4). The temperature of the room will be about 62 degrees Fahrenheit, and the weight of the basketball in this room will be about 1.2 pounds. In the middle of the room, the basketball hoop can be seen, and five spots around the room are placed for you to choose your level of difficulty for shooting. Since the game takes part on the Dark Spot of Uranus, you will be able to see a projection of the Dark Spot seen in Figure (U.2) on the floor of the room.

(U.1)



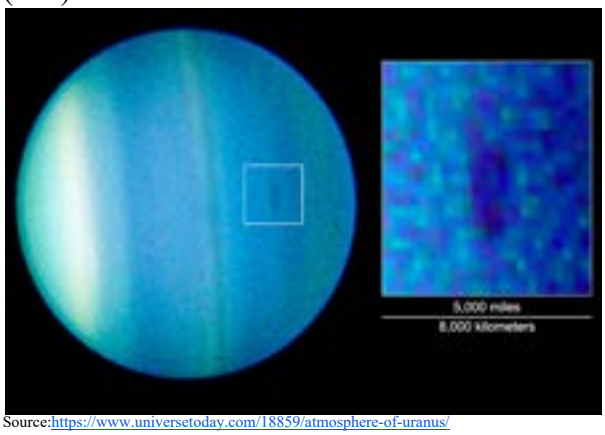
Source:https://solarstory.net/planets/uranus

(U.3)



Source: http://www.solstation.com/stars/uranus.htm

(U.2)



Source:https://www.universetoday.com/18859/atmosphere-of-uranus/

(U.4)

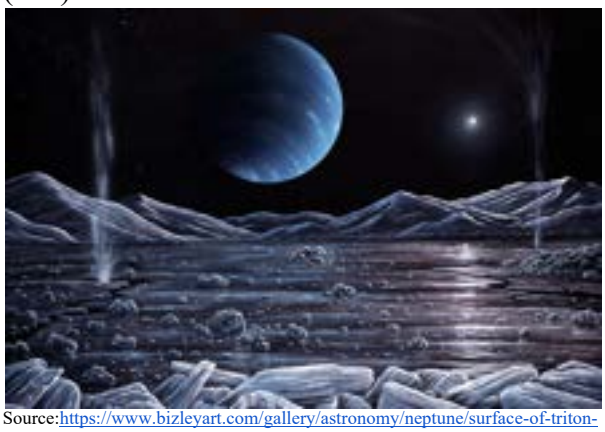


Source:https://www.shutterstock.com/video/clip-573805-cgi-render-planet-uranus-inside-planetary-rings

Neptune (Triton)

Neptune's portion of the IGBA tournament takes place on one of its moons, Triton, which will be set to 60 degrees Fahrenheit. The entire floor will be quite icy similar to Figures (N.1) and (N.3). Ice hills like the ones seen in Figure (N.1) will be surrounding the room, and three big geysers like the ones in Figures (N.1), (N.2), and (N.3) will be in the middle of the room in a triangle formation. In the middle of this triangle is where the basketball hoop will be placed, and five spots of varying difficulty will be placed around the hoop. The weight of the basketball on Triton will be about 1.6 pounds. In the back of the room, huge projection of Uranus will be seen in the nighttime sky. Small green patches of grass can be found peaking through the icy floor, but overall, the room should feel as though it has a floor made completely out of ice.

(N.1)



Source:https://www.bizleyart.com/gallery/astronomy/neptune/surface-of-triton-364

(N.3)



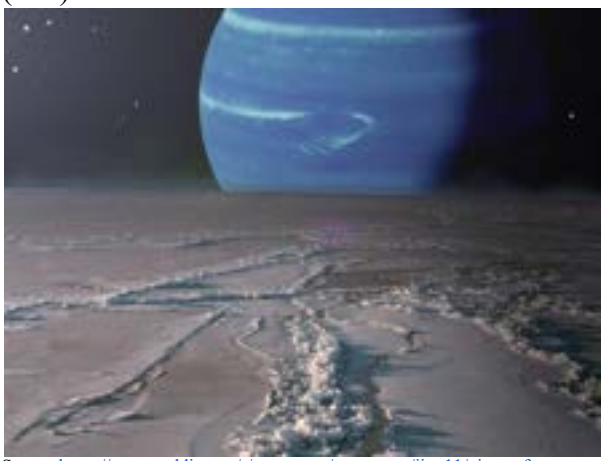
Source: https://pixels.com/featured/1-ice-volcanoes-on-triton-artwork-richard-bizley.html

(N.2)



Source:https://www.thetimenow.com/astronomy/triton.php

(N.4)



Source:https://www.reddit.com/r/spaceporn/comments/jjqu11/view_of_neptune_from_the_surface_of_its_moon/

Venus

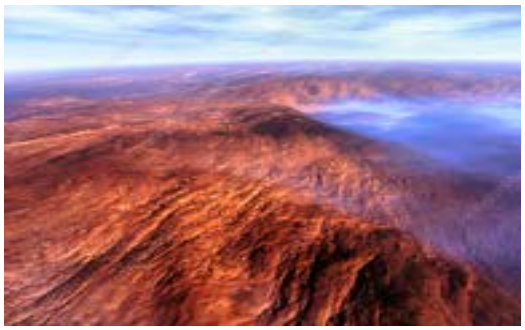
When you walk onto Venus, you are confronted with a temperature of 74 degrees Fahrenheit. The floor is a brown, dirt, sandpaper-like texture. To the right of the entrance of the room, you can see what looks like some cavernous termite mounds that are dark brown and rocky in texture. In the far back right of the room, three volcanoes can be seen, with a large one in the middle of two medium-sized ones. At the top of each volcano is lava, along with some lava at the base of the volcanoes. To the left of the room, you can see some more dark brown, tan, and dirt-colored mounds of rock. As usually, five different spots are labeled throughout the room to show where guests should shoot from. The weight of the basketball on this planet is 1.3 pounds, making the ball slightly less than the weight of Earth's basketball. There are dark clouds in the sky resulting from the eruptions of the volcanoes. Since Venus does not have any moons, no moons will be seen above. The game will take place during daytime on this planet, similar to the color of the sky seen in Figure (V.1).

(V.1)



Source:https://www.worldatlas.com/articles/are-there-volcanoes-on-other-planets.html

(V.3)



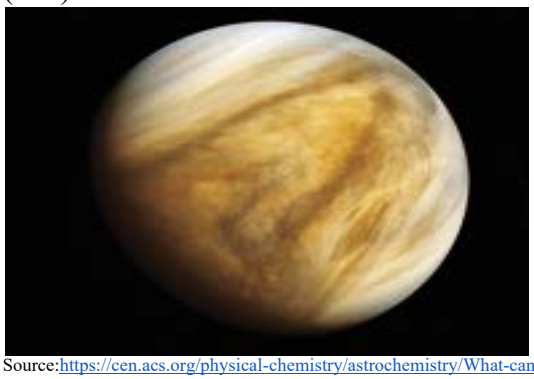
Source: https://www.sciencephoto.com/media/324998/view/coloured-3d-radar-image-of-maxwell-montes-venus

(V.2)



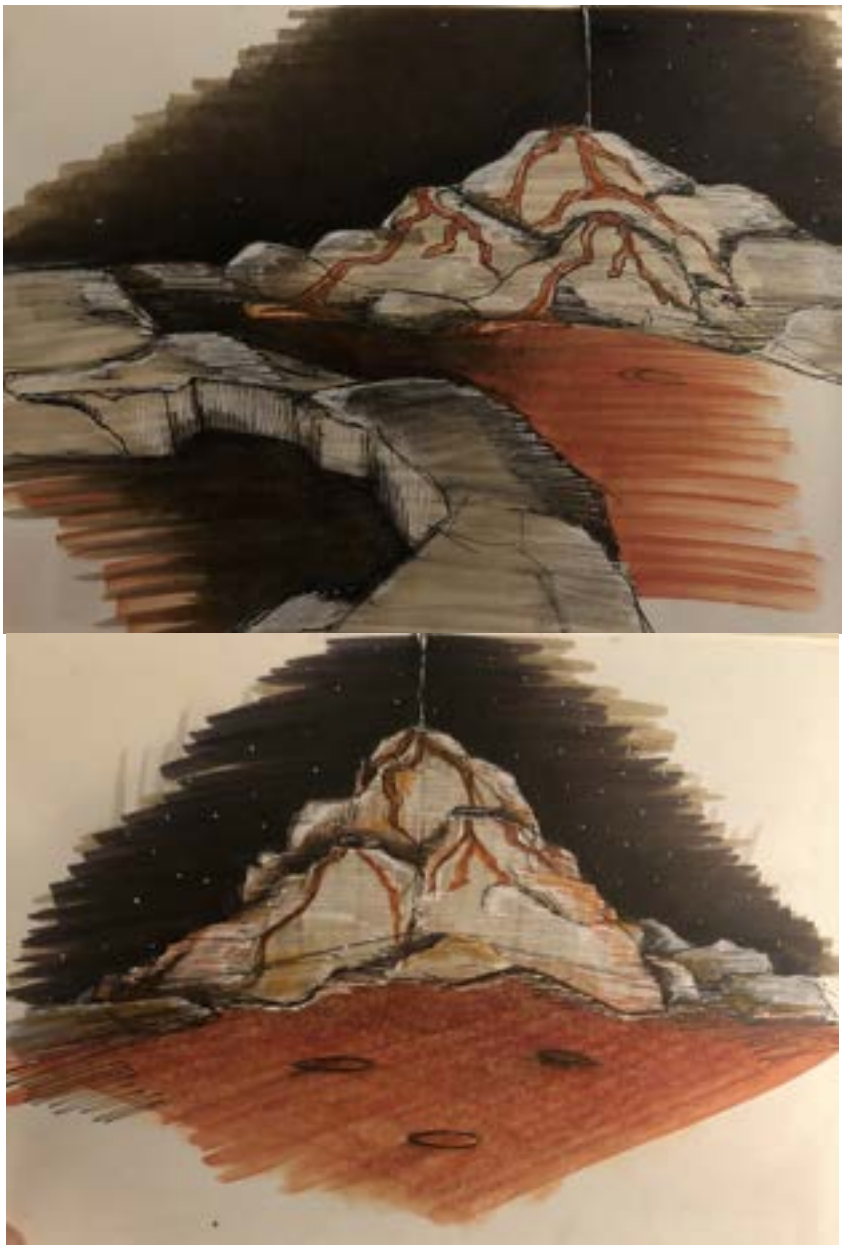
Sourc: ehttps://earthsky.org/space/hot-lava-flows-discovered-on-venus/

(V.4)

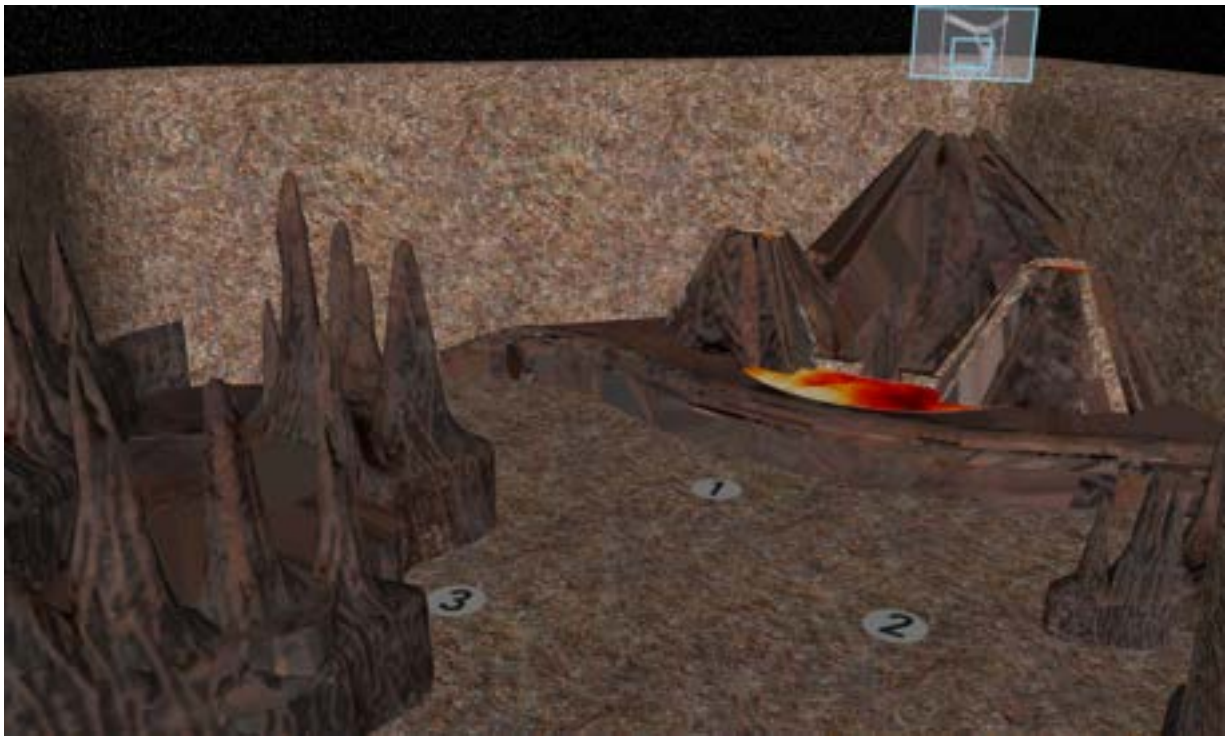


Source:https://cen.acs.org/physical-chemistry/astrochemistry/What-can-we-learn-from-Venus/99/i11

Rough Sketches of Venus



3D Model of Venus



Mercury

The surface of Mercury looks very similar to the surface of Earth moon; Mercury, however, does not have any moons. As you walk into the room housing Mercury, you will notice it feels a little warmer than the other planets at about 72 degrees Fahrenheit. You are standing right in the middle of Mercury's biggest crater, Caloris Basin, which is pictured in Figure (N.3). The room looks almost exactly like the inside of the Caloris Basin, and on the middle hill is where the basketball hoop will be placed. The weight of the basketballs on Mercury will be about 1.1 pounds. Colors similar to the metallics seen in Figure (N.2) will be used as accent colors throughout the landscape. On the left side of the room, the sun will be blazing bright as it is pretty close to Mercury. On the right side of the room, in the night sky full of stars, you will be able to see the Mariner 10 passing by on its journey through space!

(N.1)



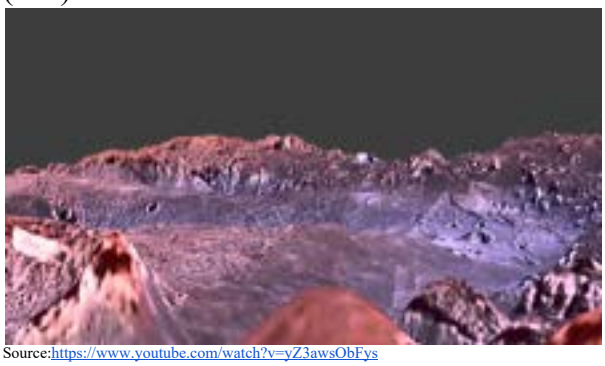
Source:https://en.wikipedia.org/wiki/Mercury_(planet)

(N.3)



Source: https://www.flickr.com/photos/gsfc/6323325370

(N.2)



Source:https://www.youtube.com/watch?v=yZ3awsObFys

(N.4)

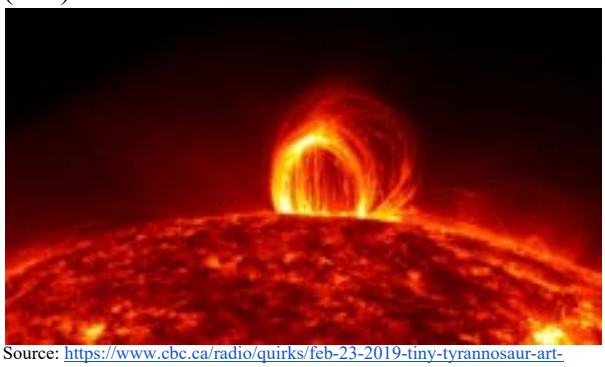


Source:
https://www.nasa.gov/feature/45-years-ago-mariner-10-off-to-venus-and-mercury/

The Sun

When you walk onto the Sun, it is immediately extremely hot in the room, and the temperature is about 80 degrees Fahrenheit. You look down on the ground to see projections of the Sun's surface, similar to what is seen in all four of the figures below. In front of you, up against the left side of the room, you see four solar flares just like the one in Figure (V.1). Two of them are larger than the other ones and are surrounded by bright whiteness on the ground to show how hot the ground is. At the opposite end of the room, you will see three volcanic-looking eruptions that almost appear like cones of lava. Since the Sun's gravity is so strong, the weight of a basketball on this planet will weigh about two pounds. The sky surrounding the room is black at the top but a light orange and the bottom followed by a brownish-red to represent the exertion of heat off of the planet and into space. In the sky, you can see very small versions of each of the planets contained within this game represented by their models hanging about and orbiting the room.

(V.1)



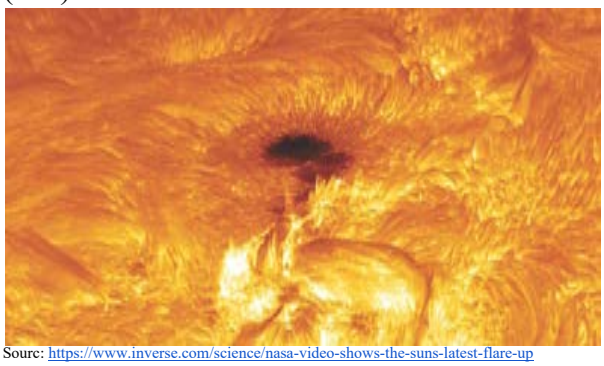
Source: https://www.cbc.ca/radio/quirks/feb-23-2019-tiny-tyrannosaur-art-acne-what-zebra-stripes-do-and-more-1.5028231/question-what-s-it-like-standing-on-the-surface-of-the-sun-1.5028238

(V.3)



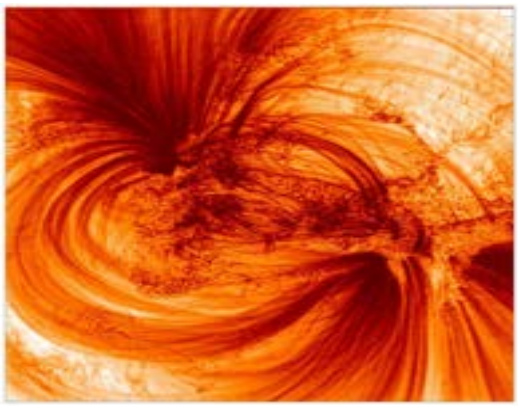
Source: https://phys.org/news/2017-08-sun-core-rotates-faster-surface.html

(V.2)



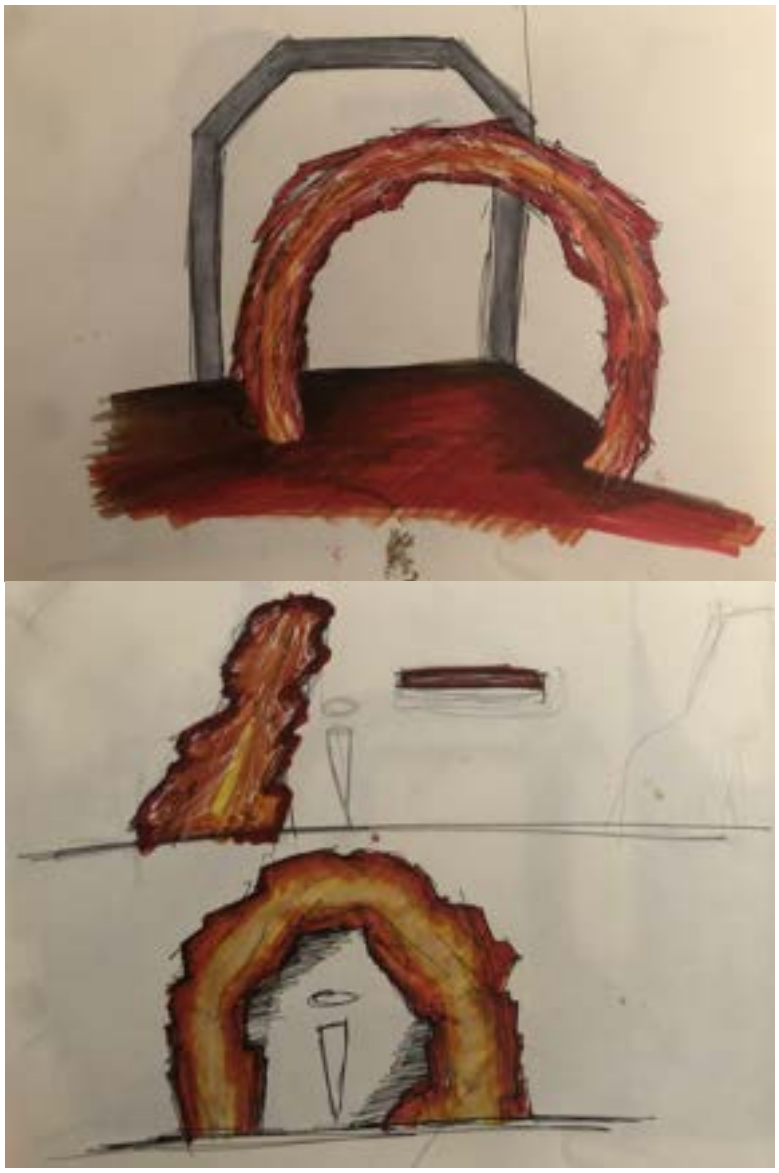
Sourc: https://www.inverse.com/science/nasa-video-shows-the-suns-latest-flare-up

(V.4)

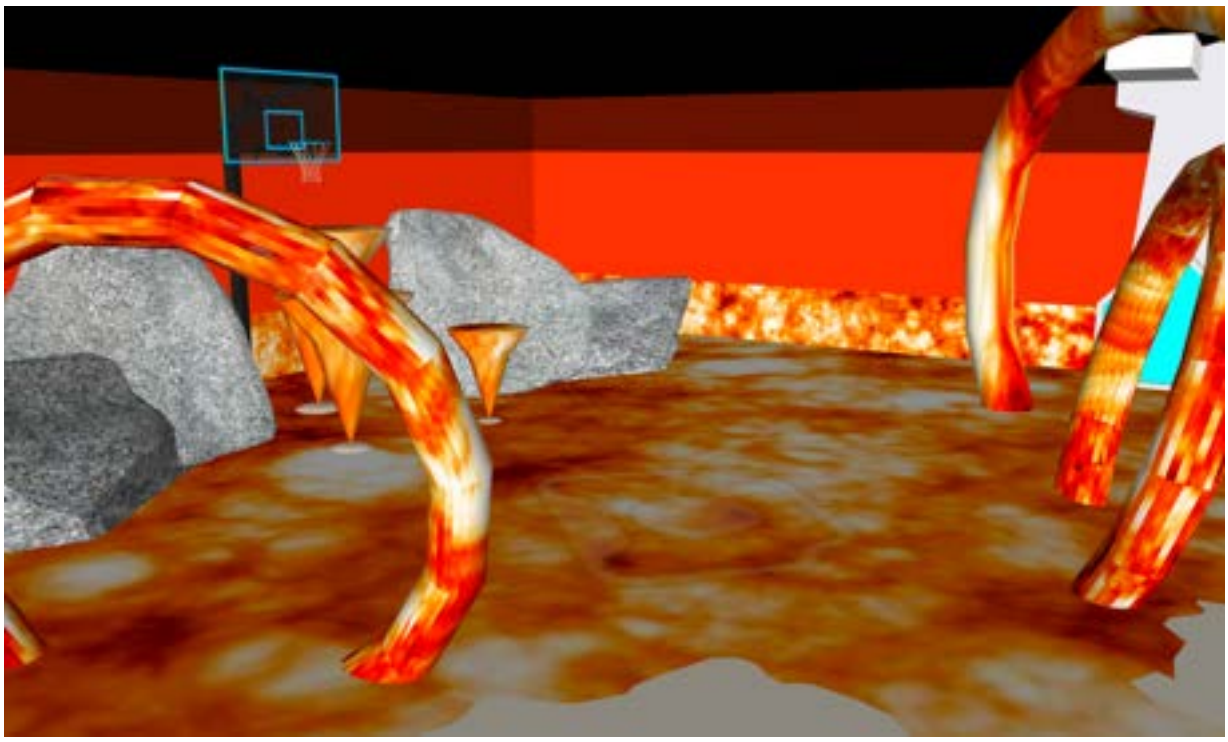


Source:https://www.blogpreston.co.uk/2020/12/nasa-2021-calendar-to-feature-picture-of-the-sun-taken-by-prestons-university-researchers/

Rough Sketches of Solar Flares



3D Model of The Sun



Earth (After the Game) & Gift Shop (Inside the House After the Game)

Once you finish the game and enter back onto Earth, you will exit through what looks like the same portal that you entered. However, this is a completely different portal that is going to put you back out to an identical recreation of Atom Silver's backyard, which will be decorated exactly the same as the previously mentioned backyard. You will then continue back into a new, completely identical recreation of Atom Silver's house in order to exit the game. In this house, you will be able to do some shopping when you encounter the gift shop.

Temperatures for the Game Rooms (Hottest to Coldest)

Sun	80 degrees Fahrenheit
Venus	74 degrees Fahrenheit
Mercury	72 degrees Fahrenheit
Earth	70 degrees Fahrenheit
Mars	68 degrees Fahrenheit
Jupiter	66 degrees Fahrenheit
Saturn	64 degrees Fahrenheit
Uranus	62 degrees Fahrenheit
Neptune	60 degrees Fahrenheit
Pluto	58 degrees Fahrenheit

Weights of the Basketballs for Each Game Room (According to Gravity)

Pluto	0.9 lbs
Mars	1.0 lbs
Mercury	1.1 lbs
Uranus	1.2 lbs
Venus	1.3 lbs
Saturn	1.4 lbs
Earth	1.5 lbs
Neptune	1.6 lbs
Jupiter	1.7 lbs
The Sun	2 lbs

Script

THE IGBA'S "SHOOTING FOR THE STARS" CHALLENGE

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Coreyclancy17@knights.ucf.edu

SCENE ONE: ATOM SILVER IN THE BACKYARD

Characters:

Atom Silver

[The video begins to play on the white sheet that is put up on the right side of Atom Silver's backyard. Space music mixed with sounds characteristic of the NBA begin to play, and Atom Silver, a tall, muscular man in a blue basketball jersey that says "Earth Citizens" with stars on it shows up on the screen. He appears to be in his childhood bedroom surrounded by small inventions of his, including the space portal that is seen in the backyard. Jerseys, signed basketballs, and trophies surround his room]

ATOM SILVER

Hello there everybody! Can you hear me? I am just right upstairs from you where I have my control panel to launch you all on your mission! What is your mission? Well, funny you should ask…you are here today to see if you qualify to be on Earth's brand-new team to the Intergalactic Basketball Association! I have been working my entire life to find a way for us Earth Citizens to qualify to play in the association.

First, I had to explain to the IGBA that since they only have nine teams, which are the Sun Flares, Mercury Messengers, Venus Magmas, Mars' Reds, Jupiter Giants, Saturn Golden Rings, Uranus Icicles, Neptune Retros, and Pluto Dwarfs, they needed a tenth in order to have an even number for tournaments. They were thinking about taking Pluto away to make an even number of eight, but after talking with the Plutonians, we knew we had to find another option, which was adding Earth! I also had to explain to the IGBA that adding Earth would bring them so many more fans and much more profit.

Once the IGBA agreed to this, all I had to do was use my astrophysicist knowledge along with my world-renowned basketball skills to create a spaceship that ALL humans can use to take to space. Once you board my ship, I will send you off into space from my upstairs control room! You will be sent to the IGBA's Space Hub, where teleportation devices to each planet will be located for you to use and prove that you have what it takes to play basketball in this galactic space league! Here's the catch though… only ONE of you will qualify in each group. Whoever gets the highest score will join our team, the Earth Citizens!

Here are the rules of the game:

> You may choose one of five numbered spots to attempt a shot from. These spots are labeled from numbers one to five, and the difficulty level increases as the numbers go up, but with that, so does the point value!
>
> You will have three attempts to shoot and sink the basket. For each made basket, you will receive the point value associated with the designated spot you shot from. If the shot is not made, no points will be awarded for the attempt. You may adjust shooting spots for each attempt.
>
> Whoever has the highest score by the end of the game will win the challenge and be inducted into the Earth Citizens team!
>
> For Junior players, the hoop MAY be lowered, and point values adjusted to make the competition fair for all ages!

Alright everyone! What do you say, are you up for challenge? If so, go on in the spaceship! I will video chat you once you are inside and send your ship off for takeoff!

[The video ends]

SCENE TWO: ATOM SILVER AND GUESTS IN THE SPACESHIP

Characters:

Atom Silver

Mars Grover

Lil Spacey

[Guests make their way into the spaceship, where they are greeted by another video screen. As soon as the door to the spaceship closes, and the video screen turns on! Atom Silver is again in his childhood home and ready to send the players to space.]

ATOM SILVER

Well, I am relieved to see I did not lose too many of you! So how about it, are you ready for blast off? Make sure you are holding on to something, because it is about to be a bumpy ride. As soon as I press my special button, you will be on your way! I have a lot more to do over on my end to make sure everything works out, so you won't hear from me until you get back, but until then, I will leave you with my "Space Center" host friends, Mars Grover and Lil Spacey! You will meet them after takeoff. But until then, shoot straight everyone! Get ready for blast off in 3…2…1… BLAST OFF! Good luck everyone!

[Atom Silver disappears from the screen. The rocket begins to shake and feels like it is blasting off into space. After a series of turbulent feelings, the rocket begins to calm down as it travels in space. Suddenly, the video turns back on again, and two reporters/ anchors show up on the screen. They are seated at a high-tech desk where they will report throughout the entire experience!]

MARS GROVER

HELLOOOOOOOO EVERYBODY! My name is MARS GROVER, and WELCOME TO THE IGBA'S "SHOOTING FOR THE

STARS" CHALLENGE on "SPACE CENTER"! I will be your host today, along with my cohost…

LIL SPACEY

Lil Spacey! That's right, we will be hosting your journey today! Right now, you are headed on your way to the Intergalactic Basketball Association's Space Hub, where they host meetings for the IGBA and allow people to easily transport from one planet to the next. In the Space Hub, keep your eyes open for artifacts of the IGBA, as well as team jerseys and flags! It looks like you are docking now though!

MARS GROVER

Yes! We will see you very soon! Enjoy the Space Hub, and we will make sure to catch up with you when you are on your way to your first planet!

[Guests exit the spaceship and wander around until they are called to begin their game. They then walk to the first elevator.]

SCENE THREE: PLUTO

Characters:

Mars Grover

Lil Spacey

[Guests enter the elevator teleporter to travel to their first planet…Pluto. On the screen inside, Mars Grover and Lil Spacey show up.]

MARS GROVER

Welcome back to Space Center everyone! You are on your way to your first planet…PLUTO! Remember, once you get there, there are five spots to shoot from. The higher the number of the spot, the more difficult it is, and thus the more points you will receive. You only get three shots though, and you may change your position at any time!

LIL SPACEY

Before you arrive though, here are some things you should know about Pluto! It is technically a dwarf star and not a planet! It can be very cold, but ice can actually turn into gas without first becoming a liquid! It can be about -387 degrees Fahrenheit, so bundle up or make sure to work your body! As with most of the locations you are going to today, there are lots of rocks and difficult terrains. Also, the gravity on Pluto is about 7% of Earth's gravity, so it may be a bit more difficult to shoot that basketball!

MARS GROVER

Exactly that Lil Spacey. It does appear, however, that we are pulling on up to Pluto! Good luck everyone. Your challenge begins now!

[The video turns off, the doors of the Elevator Transport open, and the guests go on to Pluto.]

SCENE FOUR: MARS

Characters:

Mars Grover

Lil Spacey

[After they complete their game on Pluto, they are redirected back to the elevator, where the screen turns on.]

LIL SPACEY

You did it! You completed the first game! It is time to make it to the next planet…Mars!

MARS GROVER

Hey! It's the planet that I was named after! Mars is the fourth planet of the sun and has a rocky, reddish looking terrain. Humans have actually gotten great footage from Mars using the Space Rover, which you may see throughout your time on the planet! The temperature is about negative 81 degrees Fahrenheit, and the gravity on Mars is about 37% of Earth's gravity, which again, will affect the weight of your basketball. You will be landing on the Jezero Crater, which was believed to once have water in it! Watch out though, this area is known to be inhabited by tornadoes!

LIL SPACEY

It looks like we are here! Be careful folks, and good luck! We will see you next time on Space Center.

[The screen turns off, and the guests exit the teleporter and walk onto Mars.]

SCENE FIVE: JUPITER

Characters:

Mars Grover

Lil Spacey

[After they complete their game on Mars, they are redirected back to the elevator, where the screen turns on.]

LIL SPACEY

Woah! You are back already! Congratulations on making it yet again through one more planet! Now, we are on the way to Jupiter, one of the Gas Giants!

MARS GROVER

Yes! Jupiter is made entirely out of gas, and you will be playing basketball right on what is called the Great Red Spot! Lots of fog and colors of blues, reds, oranges, and more will greet you there!

LIL SPACEY

Jupiter, the fifth planet from the sun, is about negative 234 degrees Fahrenheit, and its gravity is 2.25 times that of Earth's! This basketball is definitely going to be one of the heavier ones you use.

MARS GROVER

Here we are folks! Good luck, and we will see you all soon!

[The screen turns off, and the guests exit the teleporter and walk onto Jupiter.]

SCENE SIX: SATURN (TITAN)

Characters:

Mars Grover

Lil Spacey

[After they complete their game on Jupiter, they are redirected back to the elevator, where the screen turns on.]

LIL SPACEY

Y'all are seriously killing it. Alright, raise your hand- who has the most points? [Guest raises hand] There you go! Keep up the good work everyone!

MARS GROVER

Next up we are going to Saturn! But, instead of going to the actual planet of Saturn, you are going to its moon, Titan, where it is believed that human life could actually inhabit. Green grass and lots of water are found there. Keep your eye out and see if you can even see Saturn and it's rings yourself!

LIL SPACEY

Saturn is the sixth planet from the sun, and its moon has an average temperature of about -290 degrees Fahrenheit! Also, its gravity is about 90% of Earth's, so this ball should be a bit easier to throw!

MARS GROVER

Once again, we are arriving! Go make us proud!

[The screen turns off, and guests exit the teleporter and walk onto Titan.]

SCENE SEVEN: URANUS

Characters:

Mars Grover

Lil Spacey

[After they complete their game on Saturn, they are redirected back to the elevator, where the screen turns on.]

MARS GROVER

We are seriously so proud of you all! Great job out there on Titan! Let us head on over to Uranus! Lil Spacey, tell us a little about Uranus.

LIL SPACEY

Gladly! Uranus is the coldest planet in the solar system at about negative 371 degrees Fahrenheit! Stay warm out there! It is extremely cold and icy, full of cold fog! It is the seventh planet from the sun, and its gravity is about 90% of Earths!

MARS GROVER

It is definitely going to be a cold one, but luckily the basketball weight should not be too different from Earths!

LIL SPACEY

You have got that right! But, enough chatter, time for them to disembark and go out there!

[The screen turns off, and guests exit the teleporter and walk onto Uranus.]

SCENE EIGHT: NEPTUNE (TRITON)

Characters:

Mars Grover

Lil Spacey

[After they complete their game on Uranus, they are redirected back to the elevator, where the screen turns on.]

LIL SPACEY

Yes, everybody yes! You are doing it! I cannot believe it! We are moving on to the sixth planet! We are about halfway done! Right now, we are going to Neptune's moon of Triton!

MARS GROVER

Neptune is the furthest planet from the son, found eight planets away! The temperature is about negative 373 degrees Fahrenheit, and on Triton, the terrain is incredibly icy with spouts of ice and crystals coming out of the ground! Every now and then, you can see some small patches of grass, but for the most part, it is all covered in ice!

LIL SPACEY

The gravity on Triton is about 1.1% of Earth's, so enjoy that basketball out there! We are coming in for landing. Good luck everyone!

[The screen turns off, and guests exit the teleporter and walk onto Triton.]

SCENE NINE: VENUS

Characters:

Mars Grover

Lil Spacey

[After they complete their game on Triton, they are redirected back to the elevator, where the screen turns on.]

MARS GROVER

COME IN EVERYONE! Get some warmth in here! We really are so proud of your progress. Let's get you to a planet that is a little bit warmer…Venus!

LIL SPACEY

Venus is the second planet from the sun and feels about 900 degrees Fahrenheit! The gravity on Venus is again about 90% of Earth's gravity but watch out for all of the volcanoes on the planet! Lava and magma will be everywhere!

MARS GROVER

If anything happens to you though, do not worry, we will be in to save you! Once again, we have faith in you! You have got this!

[The screen turns off, and guests exit the teleporter and walk onto Venus.]

SCENE TEN: MERCURY

Characters:

Mars Grover

Lil Spacey

[After they complete their game on Venus, they are redirected back to the elevator, where the screen turns on.]

MARS GROVER

WOW IS IT HOT IN HERE! And it is only going to stay hot…because we are on our way to Mercury! Mercury is the smallest planet in the solar system and the closest to the sun! It is going to be about 800 degrees Fahrenheit!

LIL SPACEY

Yes! Now, the surface of Mercury might look quite similar, since Earth's moon has a very similar look, with lots of craters and grey land. You are going to be landing on the crater named Caloris Basin! With a gravity about 38% of Earth's, this basketball is going to be light. We believe in you though! It appears as though we have arrived! Good luck team!

[The screen turns off, and guests exit the teleporter and walk onto Mercury.]

SCENE ELEVEN: THE SUN

Characters:

Mars Grover

Lil Spacey

[After they complete their game on Mercury, they are redirected back to the elevator, where the screen turns on.]

LIL SPACEY

Y'ALL! YOU WILL NOT BELIEVE IT! It is time for you to compete on the last planet… the SUN!

MARS GROVER

Now this is going to be a difficult one. The sun is EXTREMELY HOT. Also, its gravity is 28 times that of Earth, so this basketball is going to be HEAVY. But we do not want to waste your time anymore. You have been training for this moment, and it is finally here! GOOD LUCK!

[The screen turns off, and guests exit the teleporter and walk onto the Sun.]

SCENE TWELVE: TRANSPORT BACK TO EARTH

Characters:

Mars Grover

Lil Spacey

[After they complete their game on the Sun, they are redirected back to the elevator, where the screen turns on.]

LIL SPACEY

OH MY GOD! YOU DID IT! I had full faith in you! And…it looks like _____ has the highest score! Congratulations! You are going to be able to play on the Earth's Citizens Team!

MARS GROVER

I AM SO PROUD OF YOU ALL! That was an extreme feat that you all conquered. You have no idea how much we were rooting for you all to win…and you did it! Regardless, if you won or not, you played basketball on all of the planets, and that is AMAZING! Now, let's get you back down to Earth! It has been a true pleasure being with you all.

MARS GROVER AND LIL SPACEY

Mars Grover and Lil Spacey signing off! Thank you for joining us for today's episode of "Space Center!"

[The screen turns off, and guests exit the teleporter and walk back onto Earth.]

SCENE THIRTEEN: ATOM SILVER IN THE BACKYARD

Characters:

Atom Silver

[The guests walk back onto a new recreation of Atom Silver's yard, which looks the exact same as the one they entered. They are redirected to the white sheet where they watched Atom Silver the first time in the yard. Atom Silver comes on the s

ATOM SILVER

CONGRATULATIONS EVERYONE! YOU DID IT! I am so proud of you all! I am especially proud of the winner of the highest points! You are now inducted into the team, "Earth's Citizens." But…I have some more news…so that everyone can have a chance again, you will all get 25% off of your next visit so that you may try one more time to join the team! You all have been an incredible joy! Now, go to my gift shop and buy some things so that I can fly us to a different GALAXY next time! My name is Atom Silver, and it has been a true pleasure.

[The video stops, and guests are redirected to the recreation of the house that now serves as a gift shop.]

THE END

Research

Earth

- Only planet known to maintain life
- Approx. 4.5 billion years ago, the earth formed from particles left over from the creation of the sun
- Solid inner core with a liquid outer core and a surrounding mantle and crust
- 3rd planet from the sun (93 million miles from the sun)
- Water has been around since the beginning of the Earth's creation and can be in any of its forms
- Only known place with LIQUID water on the surface, responsible for hydrating the land and collecting and pooling to form oceans and fresh water systems
- About 3.8 billion years ago, primitive life existed in the form of microbial organisms which eventually led the way to more advanced life forms
- 1.5 million species of plants, animals, bacteria, and fungi

Mars

- In Roman mythology, Mars symbolized blood and war
- Created from gas and dust swirling together
- Diameter is width of Africa
- 4th planet from sun
- Dense, rocky composition made up of iron, nickel, and sulfur
- Water can be found on planet in polar ice caps, temperatures drop as low as -225 degrees fahrenheit

Jupiter

- 5th largest planet from our sun and largest planet in the solar system
- 4 large moons and smaller moons orbiting around like a mini solar system. 53 confirmed moons and 26 provisional moons that have not been confirmed quite yet
- Similar composition to the sun, full of hydrogen and helium. Due to pressure and temperature increase, the hydrogen gas compresses to a liquid. This gives Jupiter the largest ocean in the solar system
- The stripes and swirls are windy clouds of ammonia and water in an atmosphere of hydrogen and helium
- Gas giant doesn't have a true surface, a spacecraft would not have anywhere to land on Jupiter but it also can't fly through Jupiter. The extreme pressure and temperatures inside the planet would crush and vaporize any spacecraft that attempted to fly through the planet
- Named after the king of the ancient Roman gods
- One day on Jupiter is 10 hours

Saturn

- In Roman mythology, named in honor of agriculture
- 2nd Largest, 10x wide as Earth
- Planet is made of mostly helium and hydrogen, making it more buoyant than water (if a body of water large enough existed, the planet would float)

- Planet has no turn surface due to gaseous composition, surface area is very cloudy (gold color clouds)
- Human life could possibly be sustained on two of Saturn's 82 Moons (Titan, Enceldus). Both moons have underground water sources
- Encelidus is as wide as PA
- Titan is as wide as Canada, has clouds and an atmosphere
- Ring around planet is only ⅔ mile thick, but as wide as 4 earths
- Rings are made of dust and debris from comets and asteroids, particles range from the size of dust to mountains.
- 2nd smallest

Uranus

- The Blue World
- Seventh planet from the sun
- Orbits the sun once every 84 earth years (approx. length of a human's entire life): a human living on Uranus would experience the seasons once but each for about 21 years
- Coldest temperatures in the solar systems (-370 degrees Fahrenheit)
- Has the width of four Earths
- Earth sized core made of iron and magnesium silicate
- 80% of Uranus is a worldwide ocean of ices made of water, ammonia, and methane
- Icy composition prevents Uranus from emitting heat from the core
- The other planets spin on their axes like tops; Uranus rolls around its equator, and its polar regions point toward and away from the sun rather than upward or downward
- 13 rings and 27 moons; these orbit the planet vertically instead of horizontally

Neptune

- 8th planet from the sun, 30x the distance of Earth to the Sun
- 1 Neptune year = 165 Earth years
- Cold, dark, icy planet because of distance from the Sun
- Core is made of rock, ice, and water. 1.5x the size of Earth
- Abundance of frozen methane gas results in blue color
- 4th largest
- Has supersonic winds that reach as high as 1110mph, blow in retrograde

Venus

- Referred to as the sister planet to Earth due to sizing and mass
- Surface is hidden by an opaque layer of clouds due to the sulphuric acid
- Second largest terrestrial planet as well as second brightest natural object in the sky
- Hottest planet in the solar system, does not shift temp from night to day
- Ancient civilizations referred to Venus as the morning star and the evening star, not realizing it was the same planet they were seeing. When the orbit of Venus overtakes Earth's orbit, it changes from being visible at sunrise to being visible at sunset. They were known as Phosphorus and Hesperus by the Greeks, and Lucifer and Vesper by the Romans.

Mercury

- Named after the messenger of the Roman gods
- Terrestrial planet
- Smallest planet
- Huge iron core
- Closest to the sun
- 800 degrees Fahrenheit during the day, -290 degrees Fahrenheit at night
- Visited by Messenger and Mariner 10
- Lack of an atmosphere; very thin exosphere made of oxygen, sodium, hydrogen, helium, and potassium

The Sun

- Yellow dwarf star
- Electric currents generate a magnetic field that is carried throughout the solar system by the solar wind (a stream of electrically charged gas blowing outward in all directions)
- Not as large as other types of stars in the solar system

Pluto

- Very low gravity
- Extremely thin atmosphere, made out of nitrogen, some methane, and a bit of CO2
- Haze on the surface of Pluto
- On summer days, the ice will vaporize into thin air without becoming a liquid first
- Dwarf star, not an actual planet

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Certification of Group Project Contributions

Corey Clancy completed the following tasks for this project:

- Concept Development
- Story Writing
- Research (Sun, Pluto, Jupiter, Uranus)
- Treatments (The Overall Space, Earth: Inside Atom Silver's House, Earth: Atom Silver's Backyard, The Spaceship to the IGBA Hub, The IGBA Space Hub, Elevator Transports, Pluto, Mars, Jupiter, Saturn [Titan], Uranus, Neptune [Triton], Venus, Mercury, The Sun, Earth [After the Game], Gift Shop [Inside the House After the Game])
- Storyboard Displaying Game Play
- Script
- Scales
    - Heat Scale
    - Gravity Scale
- Organized the "IGBA's 'Shooting for the Stars' Challenge" Document/ Documentation

Allison Paci completed the following tasks for this project:

- Concept Development
- Research (Mercury, Earth, Venus)
- Organized the Slide Deck/ Presentation Layout
- Sketching/ Illustrations/ Visuals

Ryan Cassidy completed the following tasks for this project:

- Concept Development
- Story Writing
- Research (Mars, Saturn, Neptune)
- Game Play Development
- Modeling/Visuals
- Coordination